

Emilie PERNET

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WORK EXPERIENCE

PhD Student

01/10/2024 – Present

Institut d'Astrophysique de Paris

Paris, France

Supervisor: Dr. Stéphane Charlot

- “Spectral modeling of early galaxies integrating binary-star processes”
- Building a grid of photoionised models with CLOUDY, using GALSEVN binary-star SSP models, to implement in the BEAGLE SED fitting tool.

Doctoral Mission

01/10/2024 – Present

Académie Spatiale d'Ile de France - Sorbonne Université

Paris, France

Coordination and Organization of Space-related Events for School and Undergraduate Students

- Outreach for l'Académie Spatiale d'Ile de France
- Organized two-week internships for high-schoolers, to visit space-related labs

Student researcher

06/02/2023 – 08/12/2023

Instituto de Astrofísica de Canarias

La Laguna, Santa Cruz de Tenerife, Spain

Supervisor: Dr. Ignacio Martín Navarro

- “The evolution of galaxies through their stellar populations”
- Carried out full spectral analysis of SDSS stacked spectra, using pPXF
- Led to two publications and an oral presentation at an international conference

“Discovery” internship (stage de découverte)

October 2017

Université Grenoble Alpes

La Tronche, Isère, France

Supervisor: Mikael Martin, Lab Manager

- Discovery of a research lab environment (technicians, engineers, researchers)
- I took part in: bacterial experiments, preparation of media for bacteria and yeast, polymerase chain reaction (PCR) and agarose gels.
- I learnt: how a research team works, how to work within a team, what is to be a researcher

PUBLICATIONS

Pernet et al. 2024. *[Mg/Fe] & variable IMF: a revision of $[\alpha/\text{Fe}]$ for massive galaxies*. 2024, A&A, 687, L14

Iglesias-Navarro et al. 2024. *Deriving Star Formation Histories of Galaxies from Spectra with Simulation-based Inference*. 2024, A&A, 689, A58

CONFERENCES

Charting Cosmic Dawn in Copenhagen

April 2026, *Helsingør Denmark*

“Next-generation spectral models for high-redshift galaxies observed with JWST” (Oral presentation)

Advanced Summer School Galaxy Formation and Evolution Summer School

July 2025, *Spetses Greece*

Unveiling the physics of early galaxy and black hole formation with JWST

December 2024, *Paris France*

Baryonic Cycle Across Space & Time Winter School

October 2024, *Tenerife Spain*

A life devoted to stellar population

October 2023, *Tenerife Spain*,

“ $[\alpha/\text{Fe}] \neq [\text{Mg}/\text{Fe}]$ misleading trends for massive galaxies” (Oral presentation)

EDUCATION

Sorbonne Université
PhD

Paris, France
10/2024-Present

- Three-year PhD program, funded by the Ecole Doctorale Astronomie et Astrophysique d'Ile de France, and the Académie Spatiale d'Ile de France.

University of Surrey
Master : MPhys with Nuclear Astrophysics

Guildford, UK
09/2020-06/2024

- Master awarded with First Class Honours
- Four-year course, including 10 months of research (Feb-Dec 2023)
- Year 1: 82.6% ; Year 2: 82.8% ; Year 3: 78% ; Year 4: 74.3%
- Modules including, in particular: Quantum Physics, Atoms to Lasers, Special Relativity, General Relativity, Nuclear Astrophysics, Explosive Stellar Phenomena, Modern Computational Techniques, Research Techniques in Astronomy, Principles of Data Science
- Awards: 'The NPL MPhys prize in Physics' for the best overall MPhys performance at FHEQ Level 7, 'The Department of Physics Level 5 Prize' for the best overall performance at Level 5 of the Physics UG Programmes and 'The Dr. Chris Sullivan Prize': for the best performing second-year student in Atoms to Lasers

CNED (online school) & ASPEN (private structure)
French Baccalauréat, Scientific

Meylan, France
2018-2020

- Final grade (weighted average): 16.88/20, "Mention très bien" (with highest honour)
- Detailed grades: Physics-Chemistry: 20/20, Mathematics: 19/20, Biology: 17/20, History: 17/20, English: 14/20, Philosophy: 14/20, German: 14/20, French: 12.5/20
- I completed the three years of high school in two years

SKILLS

Computing

- Fluent in Python, main libraries (e.g., NumPy, Matplotlib) and specific libraries (e.g., pPXF, astropy, pandas)
- Good knowledge of C and Fortran (e.g., CLOUDY and BEAGLE)
- Regular usage and good understanding of clusters, managed under SLURM and HTCondor
- Competent with Microsoft 365, especially Excel, used for data analysis
- Competent with Latex/Overleaf, used to write reports, papers, and other documents

Communication and team work

- Active member of the RPS working group. Worked as PhD representative to identify and assess the psychosocial risks at the IAP
- Active member of the OSU-IAP council
- Coordination of weekly coffee between speakers and PhD students
- Served as class representative for 3 years, facilitating communication between students and teachers, as well as active member of the Administration Board in secondary school
- Led university group projects, including studies on white dwarfs, asteroid trajectory and interception, and experiments with a Fabry-Perot Interferometer
- Demonstrated communication skills within a research team through active participation in group meetings, journal clubs, and related activities

REFERENCES

Name	Employer	Email
Stéphane Charlot	Institut d'Astrophysique de Paris	charlot@iap.fr
Ignacio Martín Navarro	Instituto de Astrofísica de Canarias	imartin@iac.es
Lisa Morrison	Physics Department, University of Surrey	l.n.morrison@surrey.ac.uk